

Pennies for Pads: Subsidized Menstrual Health Products and Adoption in India*

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Abstract

We estimate the effect of India's *Suvidha* program, which sells sanitary napkins at ₹1 per pad through *Jan Aushadhi Kendra* pharmacies. Linking administrative records to two National Family Health Survey rounds, we apply a doubly robust difference-in-differences estimator to women aged 15–24. Subsidized availability raises the use of locally prepared or sanitary napkins by 9.6 percentage points, a 16% gain over a roughly 60% comparison mean. A falsification test using *Kendra* openings before subsidized pads existed is consistent with the subsidized product, rather than *Kendra* presence per se, as the operative channel. Effects are larger where a *Kendra* opens near hospitals or supermarkets and at woman-owned outlets.

Keywords: menstrual health, sanitary napkins, price subsidies, health product adoption

JEL Codes: I12, I15, I18, H51, J16

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1 Introduction

Around the world, governments have made menstrual hygiene products a focus of public policy. Over the past two decades, they have moved on two fronts to make commercial sanitary napkins more affordable: removing sales and value-added taxes on these products, and distributing them free or at subsidized prices.¹ These policies share a single premise—that lowering the cost of menstrual products will raise use—yet whether subsidizing sanitary napkins actually increases their adoption has attracted remarkably little direct evidence. This paper provides such evidence: we estimate the effect of deeply subsidized sanitary napkins delivered through India’s *Jan Aushadhi–Suvidha* program.

Closing gaps in menstrual product use is widely viewed as consequential, and the most recent causal estimates are large. Free sanitary-pad dispensers in schools in India lowered girls’ dropout and raised their attendance by 24 percentage points ([Agarwal, Chia, and Ghosh, 2024](#)); free pads cut school absenteeism by roughly half in Kenya ([Benshaul-Tolonen, Zulaika, Nyothach, Oduor, Mason, Obor, Alexander, Laserson, and Phillips-Howard, 2021](#)); a school program pairing products with stigma reduction raised learning by 0.15 standard deviations in Madagascar ([Macours, Vera Rueda, and Webb, 2024](#)); and disposable pads cut women’s work absenteeism by about 24 percentage points in Burkina Faso ([Krenz and Strulik, 2021](#)). Product access is also linked to better reproductive health ([Phillips-Howard et al., 2016](#)).

Affordability remains a potential barrier to adoption: a 2019 account describes the difficulty low-income households in India face in paying for menstrual products ([Melnik, 2019](#)), while non-pecuniary stigma around menstruation can compound the barrier ([Sivakami et al., 2019](#)). Roughly two-thirds of Indian women aged 15–24 report using cloth, and pad-use gaps between above- and below-median wealth households exceed thirty percentage points in NFHS-4 ([International Institute for Population Sciences and ICF, 2017](#)). This paper studies whether subsidizing sanitary

¹Kenya exempted sanitary towels and tampons from VAT in 2004 ([International Monetary Fund, 2009](#)). The United Kingdom removed VAT on menstrual products in January 2021; England’s free school- and college-distribution scheme began in January 2020 ([HM Treasury, 2021](#)). Scotland’s statutory free-provision duties took effect in August 2022 ([Scottish Government, 2022](#)). New Zealand announced a school rollout from June 2021, while France’s 2021 initiative covered university students ([New Zealand Government, 2021](#); [French Government, 2021](#)). In India, the GST Council recommended exempting sanitary napkins in July 2018 ([Ministry of Finance, 2018](#)); the government approved the Menstrual Hygiene Policy for School-Going Girls in November 2024 ([Ministry of Health and Family Welfare, 2024](#)). In January 2026, the Supreme Court recognized menstrual health as part of Article 21 and directed free sanitary-napkin provision in government and private schools ([Supreme Court of India, 2026](#), paras. 172(d), 173(II)(i)).

napkins and providing them through a network of generic pharmacies increases their adoption. We examine this question through the *Jan Aushadhi–Suvidha* program, which sells subsidized *Suvidha* sanitary napkins at ₹1 per pad—against a prevailing commercial price of ₹3–8—through the government’s nationwide *Jan Aushadhi Kendra* generic-pharmacy network ([Press Information Bureau, 2019](#)).

The *Jan Aushadhi–Suvidha* program differs from India’s earlier menstrual-health interventions in its delivery channel. Earlier programs relied on schools and health workers, with eligibility varying across schemes: school distribution targeted enrolled girls, while the Menstrual Hygiene Scheme targeted rural adolescent girls and RKSK explicitly included adolescents both in and out of school ([National Health Mission, n.d.b,n](#)). *Jan Aushadhi Kendras*, by contrast, sell to walk-in customers through a commercial-retail channel. This channel can serve out-of-school adolescents and adult women without school-enrollment or adolescent-program eligibility requirements. The setting therefore offers a rare opportunity to estimate the adoption response to deeply subsidized menstrual products delivered through existing commercial retail infrastructure.

We assemble a novel dataset linking administrative records on the universe of *Jan Aushadhi Kendras*—establishment dates, precise geolocations, and owner characteristics—with the National Family Health Survey (NFHS). The linked survey data contain around 500,000 respondents aged 15–24 across two rounds of the NFHS. We estimate the adoption effect with the doubly robust difference-in-differences estimator of [Sant’Anna and Zhao \(2020\)](#), conditioning on the observable characteristics that predict *Kendra* placement.

Our central finding is that access to subsidized napkins through *Kendras* raises pad use by 9.6 percentage points, a 16% increase over a roughly 60% comparison mean. Effects are larger where *Kendras* open near hospitals or supermarkets and at woman-owned outlets, and adoption builds over the first several months of availability rather than appearing immediately. Among women who are not pregnant at interview, our analysis also suggests reductions in genital sores and discharge.

These conclusions withstand a battery of robustness checks. First, because the *Kendra* network long predates the subsidized product, we conduct a falsification test: *Kendra* openings in the years before subsidized pads existed produce no adoption response, consistent with the subsidized product rather than the channels—health information, retail density, or correlated local development—through which pharmacy presence might independently have raised pad use. Second, the treatment effect estimate is positive when states running concurrent sanitary-napkin programs are excluded one at a time and when all such states are dropped together. Third, a [Rambachan and Roth \(2023\)](#) sensitivity analysis shows that the effect persists under sizeable violations of parallel trends.

The paper contributes to four strands of literature. Most directly, it speaks to a growing policy and research interest in subsidizing menstrual products. A large experimental literature provides

products directly and traces their effects on schooling, health, and labor, with notably mixed results. In Nepal, [Oster and Thornton \(2011\)](#) find that menstruation costs girls under half a day of school per year and that free sanitary products do not close that small gap, and the Kenyan feasibility trial of [Phillips-Howard, Nyothach, ter Kuile, Omoto, Wang, Zeh, Onyango, Mason, Alexander, Odhiambo, Eleveld, Mohammed, van Eijk, Edwards, Vulule, Faragher, and Laserson \(2016\)](#) likewise detects no effect of cups or pads on school dropout; where effects do emerge, they tend to run through attendance and health, as in the sizable reductions in menstruation-related absenteeism that [Benshaul-Tolonen, Zulaika, Nyothach, Oduor, Mason, Obor, Alexander, Laserson, and Phillips-Howard \(2021\)](#) estimate from pad provision in western Kenya, the attendance gains reported by the quasi-randomized pad-and-puberty-education trial of [Montgomery, Hennegan, Dolan, Wu, Steinfeld, and Scott \(2016\)](#) in Uganda, and the lower dropout that [Agarwal, Chia, and Ghosh \(2024\)](#) link to school pad provision.

Second, a complementary body of work shifts attention from access to information and social norms: [Castro and Mang \(2024\)](#) show that structured group discussions reduce menstrual stigma and raise both willingness to pay for and adoption of menstrual products; [Czura, Menzel, and Miotto \(2024\)](#) find among Bangladeshi garment workers that free pads and hygiene information improve menstrual practices and health; and [Macours, Vera Rueda, and Webb \(2024\)](#), in a field experiment across 140 Malagasy schools, show that bundling sanitation, products, and teacher-led sensitization shifts hygiene behavior and reduces stigma. We build on this work by estimating the adoption response to subsidized provision through an existing commercial-retail channel at national scale.

Third, far less is known about the question these studies presuppose—whether lowering the price of menstrual products raises their use. The closest causal evidence comes from “tampon tax” reforms in high-income countries, where studies using retail and household scanner data find that VAT cuts pass through fully—often more than fully—into consumer prices, but that demand is price-inelastic: quantities move little, and where households respond they tend to trade up to higher-quality products rather than buy more, with lower-income households gaining most from the price decline ([Büttner, Hechtner, and Madzharova, 2025](#); [Frey and Haucap, 2024](#); [Kinnl and Wohak, 2026](#)). Yet these studies observe aggregate market sales in settings where pad use is already near-universal, leaving open whether deeply subsidized provision draws in prior non-users—the margin we estimate in a setting where baseline use is far from universal.

Fourth, and more broadly, the paper speaks to the provision and pricing of inexpensive health products. Demand for such goods is typically highly price-sensitive, with even small fees sharply cutting take-up of one-off or preventive products like bednets and water treatment ([Cohen and Dupas, 2010](#); [Kremer and Miguel, 2007](#); [Ashraf, Berry, and Shapiro, 2010](#); [Dupas, 2014](#); [Dupas and Miguel, 2017](#)); menstrual products extend the question to a recurring necessity, where the relevant

margin is sustained repeat use. Because a program’s delivery channel also shapes its reach (Muralidharan, Niehaus, and Sukhtankar, 2016; Ashraf, Bandiera, and Jack, 2014; Duflo, Dupas, and Kremer, 2015; Finkelstein and Notowidigdo, 2019), our setting is doubly distinctive: rather than distributing products through schools or frontline health workers under program-specific eligibility rules, the subsidy travels through a commercial-retail network serving any walk-in customer at national scale. Retail access is not conditional on school enrollment or eligibility for an age-targeted adolescent program.

These findings carry important policy implications. Subsidized availability raises adoption, but the response is larger where the surrounding context lowers the non-pecuniary cost of obtaining a pad—where a *Kendra* opens near hospitals or supermarkets and at woman-owned outlets. These patterns show where the program has larger effects; they do not cleanly separate affordability, information, transaction costs, stigma, inventory, and retail access. They also create an efficiency–equity tension for further expansion: adoption responds most strongly in better-served areas, while unmet need is greatest elsewhere. Routed through a pre-existing commercial-retail network, the program reaches women who are not served by school-based distribution without adding to the load on schools or frontline staff.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting; Section 3 the data; Section 4 the empirical strategy; Section 5 presents the results, robustness checks, and health outcomes; and Section 6 concludes.

2 Institutional Background

India has pursued menstrual-hygiene policy through several channels over the past fifteen years. We situate the *Jan Aushadhi–Suvidha* program against the earlier school- and health-worker-based programs it departs from, the generic-pharmacy network through which it is delivered, and the contemporaneous state schemes that operated alongside it.

2.1 Earlier Menstrual-Hygiene Programs

The central government’s Menstrual Hygiene Scheme, introduced in 2011, targeted rural girls aged 10–19 and distributed subsidized sanitary napkins through Accredited Social Health Activists (ASHAs) (National Health Mission, n.d.b). The broader *Rashtriya Kishor Swasthya Karyakram* (RKSK), launched in 2014, explicitly included adolescents both in and out of school (National Health Mission, n.d.a; Desai, 2017). School-based state schemes, discussed below, used a different eligibility rule centered on enrollment.

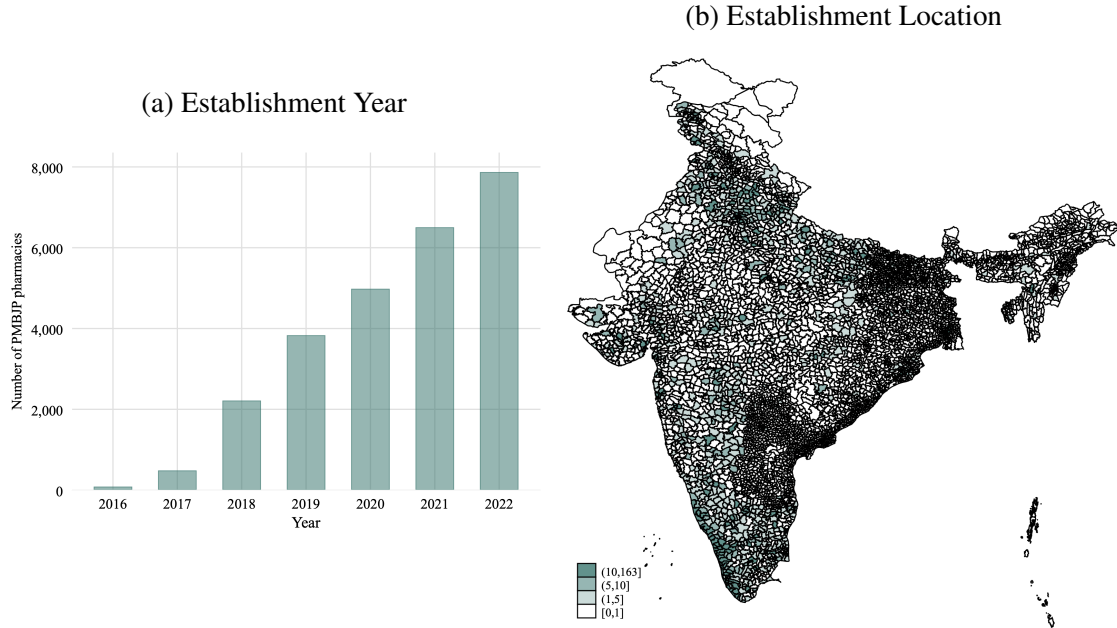
Formal eligibility does not establish effective reach. A rapid assessment in two Assam districts documents incomplete receipt of sanitary napkins among surveyed adolescent girls (10to19: Dasra Adolescents Collaborative, n.d.). This local evidence points to gaps in effective access. *Suvidha* adds a retail channel open to any buyer alongside targeted school- and health-worker-based provision.

2.2 The *Jan Aushadhi Kendra* Network

Jan Aushadhi Kendras are government-sponsored generic pharmacies that have operated since 2008, roughly a decade before subsidized napkins joined their product line. They run on commercial principles rather than through public-sector frontline workers: operators retain a 20% margin on sales and receive enhanced support in aspirational districts and for women entrepreneurs, entrepreneurs from Scheduled Castes and Scheduled Tribes, and entrepreneurs with disabilities. These groups also pay no application fee. Because a *Kendra* serves any walk-in customer, access does not depend on school enrollment or eligibility for an adolescent-health program.

Placement is administratively governed rather than random. Operators apply to the Department of Pharmaceuticals and undergo a verification and approval process that introduces variable delays in when an outlet opens, and a distance rule requires a new *Kendra* to maintain a minimum separation—ordinarily one kilometer, 1.5 kilometers in some areas—from an existing one. Openings concentrate in more developed subdistricts, a selection pattern we document in Section 3 and address by conditioning on observed placement predictors in Section 4. The network expanded sharply over the study period (Figure I). In the NFHS-linked sample, 2,206 subdistricts are ever treated. As summarized in Appendix Table A1, the median treated subdistrict contains two *Kendras*, the mean is 3.73, and the 95th percentile is 13.

FIGURE I: Establishment Year and Location of *Jan Aushadhi Kendras*



Note: Panel (a) presents the cumulative number of *Jan Aushadhi Kendras* established by year. Panel (b) shows density across Indian subdistricts, with darker shading indicating higher concentrations.

Data source: Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers, Government of India.

2.3 The *Suvidha* Sanitary Napkin

Suvidha is a government-branded, steeply subsidized sanitary napkin sold over the counter at *Kendras* alongside other health commodities—a setting in which the purchase may carry less stigma than collection through a specialized distribution point. Appendix Table A2 lists the product portfolio; the core product, regular-size with wings, retails at ₹1 per pad (₹4 for a pack of four), between one-third and one-eighth of the prevailing commercial price of ₹3–8 per pad for comparable products (Press Information Bureau, 2020).

The deep subsidy that defines our treatment took effect on August 27, 2019, when the government cut the retail price to ₹1 per pad (Press Information Bureau, 2019). *Suvidha* had been available before this date at ₹2.50 per pad but reached consumers in relatively small volume; national sales rose sharply after the ₹1 regime began (Press Information Bureau, 2020).

2.4 Concurrent State Programs

Several states operated their own menstrual-hygiene schemes during the study window or introduced schemes later in 2021. These included Kerala's *She Pad*, Odisha's *Khushi*, Maharashtra's *Asmita Yojana*, and programs in Tamil Nadu, Karnataka, Haryana, Punjab, Rajasthan, and Andhra Pradesh. Appendix Table A3 identifies the sources for each program's timing, target population, and delivery channel. Many schemes used schools or *anganwadi* centers, but eligibility was not

uniformly restricted to enrolled girls. Because programs operating during fieldwork could affect pad use, Section 5.4.6 reports exclusions of these states individually and jointly; the broader nine-state check also includes Andhra Pradesh and Rajasthan, whose cited schemes began after the stated NFHS-5 fieldwork window.

3 Data and Measurement

Our empirical analysis integrates four primary data sources: (i) administrative records on the universe of *Jan Aushadhi Kendras*, (ii) women’s health outcomes from the National Family Health Survey (NFHS), (iii) spatial amenity measures from the Geoapify Places API, and (iv) baseline subdistrict characteristics from the Socioeconomic High-resolution Rural-Urban Geographic Platform for India (SHRUG).

3.1 Baseline Socioeconomic Characteristics

We construct pre-policy subdistrict characteristics from SHRUG ([Asher, Lunt, Matsuura, and Novosad, 2021](#)), which integrates the 2011 Population Census Abstract, 2011 Village Directory, and 2012 VIIRS nighttime lights. The female literacy rate, female labor force participation, sanitation infrastructure, school availability, and nighttime light intensity predate the *Suvidha* program and capture the development gradient along which the program expanded.

Table I documents substantial baseline heterogeneity. Sixty-eight percent of communities lack organized waste collection; the Total Sanitation Campaign covers roughly one-third of subdistricts. Female literacy averages 52%; female labor force participation, 33%. Nighttime light intensity ranges from 0 to 111 (mean 1.11), reflecting the rural-urban development divide. These covariates capture important observable dimensions of the development gradient along which *Kendras* are placed. Because placement may also reflect unobserved political, institutional, or social factors, Section 4.2 examines pre-policy differences and trends directly.

TABLE I: Descriptive Statistics of Subdistrict Characteristics (Baseline)

	(1)	(2)	(3)	(4)
	Mean	SD	Min	Max
Sanitation				
No waste collection	0.68	0.40	0.00	1.00
Community has waste disposal	0.04	0.12	0.00	1.00
Total Sanitation Campaign	0.36	0.41	0.00	1.00
School and Anganwadi				
Availability of govt middle schools	0.50	0.24	0.00	1.00
Availability of anganwadi	0.87	0.15	0.00	1.00
Night Lights				
Mean night lights	1.11	5.10	0.00	111.49
Female Literacy and LFP				
Share of literate women	0.52	0.13	0.07	0.90
Share of working women	0.33	0.14	0.04	0.79
Observations				5,993

Note: Baseline subdistrict characteristics prior to the policy. Sanitation and educational infrastructure measures from the 2011 Population Census Village Directory; female literacy and labor force participation from the 2011 Population Census Abstract; nighttime light intensity from 2012 VIIRS data.

Data source: Socioeconomic High-resolution Rural-Urban Geographic Platform for India (SHRUG).

3.2 Spatial Amenity Data

Using the Geoapify Places API, we identify amenities within five minutes’ driving distance of each *Kendra*: hospitals (potential sources of health information and referrals), non-generic pharmacies (alternative sources for menstrual products), and supermarkets (indicators of retail development). These measures enable our heterogeneity analysis of infrastructure complementarities in Section 5. Table II summarizes the share of *Kendras* that co-locate with these amenities, which we discuss below.

3.3 Administrative Data on *Jan Aushadhi Kendras*

Administrative records from the Department of Pharmaceuticals document the precise location, establishment date, and owner name for all *Jan Aushadhi Kendras*. We infer owner gender through a rule-based classification algorithm implemented using a large language model (Claude 3.5) and validated against a publicly available dataset of Indian female names (Bejda, 2015). The registry does not contain store-by-period inventories, stockout histories, product-quality measures, quantities supplied to individual *Kendras*, or a validated catchment-population denominator. Our treatment measure therefore captures program availability—whether a *Kendra* is open and subsidized pads are available through the network—rather than realized stock or individual purchase.

Table II summarizes *Kendra* characteristics: 25% are woman-owned, and infrastructure co-location is substantial (63% near hospitals, 26% near existing pharmacies, 37% near supermarkets).

TABLE II: Descriptive Statistics of *Jan Aushadhi Kendra* Characteristics

	(1)	(2)
	Mean	SD
Established before Aug 27 2019	0.54	0.50
Owner is a woman	0.25	0.43
Hospital nearby	0.63	0.48
Pharmacy nearby	0.26	0.44
Supermarket nearby	0.37	0.48
Observations		8,264

Note: Characteristics of *Jan Aushadhi Kendras* across subdistricts. Amenities measured via Geoapify Places API (5-minute driving radius). Owner gender inferred via Claude AI rule-based classification validated against [Bejda \(2015\)](#).

Data source: Department of Pharmaceuticals; Geoapify Places API.

Table III documents the selection pattern: *Suvidha* becomes available earlier in subdistricts with higher female literacy, higher nighttime lights, and lower female labor force participation (a counterintuitive pattern we interpret as reflecting India’s paradoxical female labor supply, where higher participation captures low-wage informal work rather than economic advancement). This selection motivates our conditional parallel trends approach (Section 4).

TABLE III: Correlation Between Subdistrict Characteristics and Timing of *Suvidha* Availability

	Coef	S.E.
No waste collection	60.10	38.23
Community has waste disposal	-35.94	83.61
Total Sanitation Campaign	29.37	39.64
Availability of anganwadi	-16.78	58.31
Availability of govt middle schools	44.84	44.60
Share of literate women	-12.03***	1.00
Share of working women	7.57***	0.83
Mean night lights	-12.18***	3.30
Mean	1178.01	
R sq.	0.42	
Obs	5,897	

Note: Regression of the date on which *Suvidha* first becomes available in the subdistrict on baseline subdistrict characteristics with state and district fixed effects. Subdistricts where *Suvidha* never becomes available are assigned the latest observed availability date. Positive coefficients indicate that higher values of the covariate are associated with later availability. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Data source: SHRUG; Department of Pharmaceuticals.

3.4 Health Survey Data

We use two rounds bracketing the *Suvidha* launch: NFHS-4 (January 2015–December 2016) and NFHS-5 (June 2019–May 2021). The NFHS administers menstrual hygiene modules to women aged 15–24, eliciting information on use of cloth, locally prepared napkins, sanitary napkins, tampons, and cups ([International Institute for Population Sciences and ICF, 2017, 2021](#)). Our primary outcome equals one if a woman reports using either a locally prepared pad or a sanitary napkin.

The NFHS also records, for the subset of respondents administered the relevant symptom questions, whether a woman reports genital discharge or a genital sore/ulcer during the preceding 12 months. We define an indicator for any genital symptom as reporting either symptom and retain only explicit yes/no responses. We report these outcomes for the full survey-eligible sample and separately for women who are not pregnant at interview.

Appendix Table A4 documents aggregate shifts between survey rounds. The broader napkin-use measure rose from 54.8% in NFHS-4 to 74.6% in NFHS-5; sanitary-napkin use rose from 41.8% to 63.1%. Table IV shows a steep socioeconomic gradient in product use and large differences between ever-treated and never-treated locations. Women from general-caste and wealthier households, more educated women, and urban residents are more likely to use pads. Furthermore, in the smaller survey-eligible health sample, women who use pads are less likely to report a genital sore or genital discharge during the preceding 12 months.

TABLE IV: Descriptive Statistics by Treatment Status and Pad Use (NFHS Rounds 4 and 5)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Mean	Kendra ever opens			Woman uses napkin		
		No	Yes	Difference	No	Yes	Difference
Household							
Religion: Hindu	0.74	0.75	0.74	−0.01***	0.76	0.73	−0.03***
Religion: Muslim	0.15	0.12	0.17	0.05***	0.17	0.14	−0.03***
Caste: SC	0.20	0.18	0.21	0.03***	0.20	0.19	−0.01***
Caste: ST	0.18	0.26	0.12	−0.13***	0.21	0.16	−0.05***
Caste: OBC	0.40	0.37	0.41	0.04***	0.41	0.39	−0.02***
Above-median Wealth	0.50	0.37	0.59	0.22***	0.27	0.62	0.35***
Neighborhood Urban	0.25	0.13	0.34	0.21***	0.14	0.31	0.17***
Individual							
Age: 20-24	0.49	0.49	0.50	0.01***	0.50	0.49	−0.01***
Above-median Education	0.37	0.31	0.42	0.11***	0.19	0.47	0.29***
Menstruation							
Menarche: 13 or later	0.81	0.79	0.82	0.03***	0.81	0.81	0.00
Uses locally prepared or sanitary napkins	0.65	0.59	0.68	0.09***			
Kendra ever opens	0.58				0.52	0.62	0.10***
Health outcomes							
Any genital symptom	0.04	0.05	0.04	−0.01***	0.05	0.04	−0.01***
Genital discharge	0.04	0.04	0.04	−0.01***	0.05	0.03	−0.01***
Genital sore/ulcer	0.01	0.02	0.01	−0.00***	0.02	0.01	−0.01***
Observations							489,013

Note: Columns 2–4 split the sample by whether a *Kendra* ever opens in the subdistrict; columns 5–7 by whether the woman uses either a locally prepared or sanitary napkin. Difference columns report the coefficient from a regression of the row variable on the split indicator, with robust standard errors; *, **, and *** denote significance at the 10%, 5%, and 1% levels. Health-outcome rows use only respondents with valid yes/no responses to the relevant symptom questions and therefore contain 78,806–78,838 observations; all other rows use the full descriptive sample of 489,013 women. Cells are left blank where the row variable is itself the column-split variable.

Data source: NFHS Rounds 4 and 5; Department of Pharmaceuticals; SHRUG.

4 Empirical Strategy

We exploit two sources of variation: the national date on which deeply subsidized *Suvidha* pads become available and the timing of *Kendra* openings across subdistricts. In subdistricts where a *Kendra* was already open, treatment begins on the national availability date; where a *Kendra* opens later, treatment begins on the local opening date. In the linked NFHS sample, 1,705 of 2,206 treated subdistricts (77.3%) are treated on the national availability date and 501 are treated later. Fieldwork within NFHS-4 and NFHS-5 is spread across multiple years, so women are observed both before and after local availability. Appendix Figure A1 shows the distribution of interview timing relative to both *Kendra* establishment and local *Suvidha* availability.

Our main estimate uses the doubly robust difference-in-differences estimator of [Sant’Anna and Zhao \(2020\)](#) for repeated cross-sections, conditioning on the covariates that predict *Kendra* placement.

4.1 Specification

Consider woman i in subdistrict s surveyed at time t , and let $Y_{ist} \in \{0, 1\}$ denote pad use as defined in Section 3.4. The treatment indicator D_s equals one if a *Kendra* ever opens in s ; the post indicator T_{ist} equals one for interviews conducted after deeply subsidized pads become available locally—in treated subdistricts, after both a *Kendra* has opened and the ₹1 price has taken effect; in never-treated subdistricts, after the date when *Suvidha* becomes available nationally. A woman interviewed before local availability is coded pre-treatment regardless of *Kendra* presence. Our parameter of interest is the average treatment effect on the treated,

$$\tau = \mathbb{E}[Y_{i,\text{post}}(1) - Y_{i,\text{post}}(0) \mid D_s = 1]. \quad (1)$$

We estimate τ with the doubly robust estimator of [Sant’Anna and Zhao \(2020\)](#) for repeated cross-sections, which combines a propensity-score model and outcome regressions, each conditioning on a covariate vector X_{ist} : individual demographics (religion, caste, urban residence, age), district and interview month-by-year fixed effects, and baseline subdistrict characteristics (sanitation infrastructure, *anganwadi* and government-school availability, female literacy and workforce shares, and nighttime lights). The propensity score

$$\Pr(D_s = 1 \mid X_{ist}) = \frac{\exp(X'_{ist}\gamma)}{1 + \exp(X'_{ist}\gamma)} \quad (2)$$

is estimated by inverse probability tilting ([Graham, de Xavier Pinto, and Egel, 2012](#)) and the outcome regressions by weighted least squares. These enter the doubly robust moment for repeated cross-sections,

$$\begin{aligned} \hat{\tau}^{\text{DR}} = & \mathbb{E}_n \left[\frac{T}{\mathbb{E}_n[T]} \cdot \frac{D}{\mathbb{E}_n[D]} (Y - \hat{\mu}_{0,1}(X)) \right] - \mathbb{E}_n \left[\frac{1-T}{\mathbb{E}_n[1-T]} \cdot \frac{D}{\mathbb{E}_n[D]} (Y - \hat{\mu}_{0,0}(X)) \right] \\ & - \mathbb{E}_n \left[\frac{T}{\mathbb{E}_n[T]} \cdot \frac{\hat{p}(X)(1-D)}{(1 - \hat{p}(X)) \mathbb{E}_n \left[\frac{\hat{p}(X)(1-D)}{1 - \hat{p}(X)} \right]} (Y - \hat{\mu}_{0,1}(X)) \right] \\ & + \mathbb{E}_n \left[\frac{1-T}{\mathbb{E}_n[1-T]} \cdot \frac{\hat{p}(X)(1-D)}{(1 - \hat{p}(X)) \mathbb{E}_n \left[\frac{\hat{p}(X)(1-D)}{1 - \hat{p}(X)} \right]} (Y - \hat{\mu}_{0,0}(X)) \right]. \end{aligned} \quad (3)$$

The specification includes district fixed effects but not subdistrict fixed effects. Treatment status D_s is defined at the subdistrict level, so including a full set of subdistrict indicators in the propensity-score model would perfectly predict treatment and collapse the overlap needed for inverse-probability weighting. District fixed effects absorb broader geographic differences while preserving comparisons between treated and untreated subdistricts within districts. Inter-

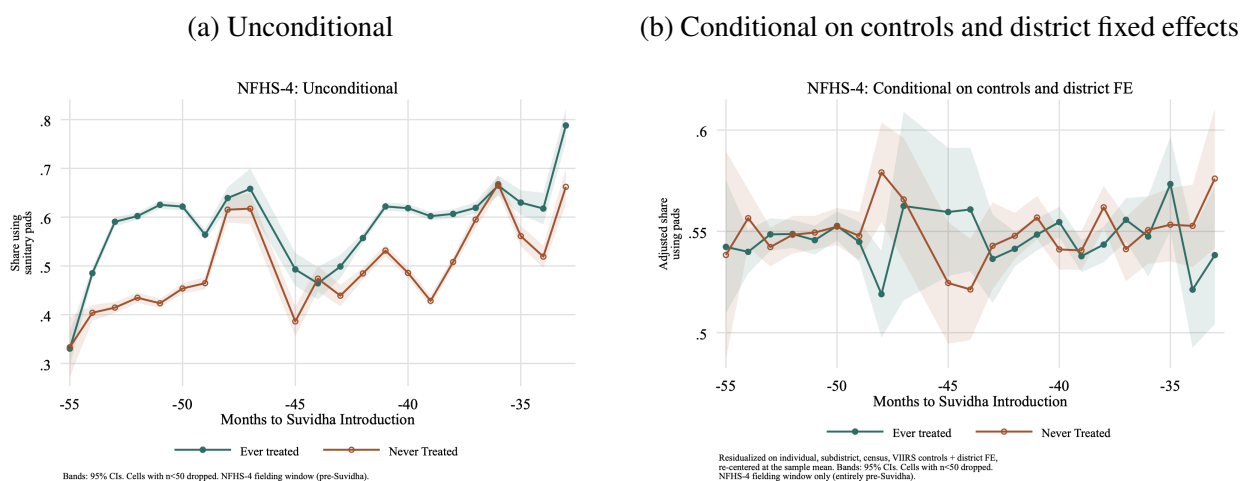
view month-by-year fixed effects absorb common interview-time shocks and are particularly useful because the COVID-19 suspension of NFHS-5 fieldwork altered the distribution of interview dates (Section 5.4.4).

4.2 Identification

Identification rests on conditional parallel trends: absent *Suvidha*, pad use in subdistricts that gain access would have followed the same path as in subdistricts that do not, conditional on X_{ist} . *Kendra* placement is not random. Operators apply to the Department of Pharmaceuticals, and ever-treated subdistricts are more urban, more educated, and more economically developed than never-treated subdistricts before the policy (Appendix Table A5). The doubly robust estimator conditions on the observed characteristics that predict placement.

The pre-policy trend and falsification tests below support the conditional parallel-trends design while recognizing that *Kendra* placement is not random. First, Figure II plots pad-use trends during NFHS-4, entirely before *Suvidha*. A formal test of the differential conditional slope yields $p = 0.838$ for pad use and $p = 0.459$ for the NFHS sanitary-napkin category. Second, the *Kendra* network predates the subsidized product by roughly a decade, which allows a falsification exercise based on *Kendra* openings before *Suvidha* existed. Section 5.4 shows no comparable adoption response around those openings.

FIGURE II: Pre-*Suvidha* Napkin-Use Trends by Treatment Status, NFHS-4 Fielding Window



Note: Both panels plot the share of women aged 15–24 using either locally prepared or sanitary napkins against months to *Suvidha* introduction (the ₹1 date), separately for subdistricts that ever versus never receive a *Kendra*, using only NFHS-4 interviews. Panel (a) plots raw shares. Panel (b) residualizes pad use on the individual and baseline subdistrict controls described in Section 4 and district fixed effects, re-centered at the sample mean. Cells with fewer than 50 observations are dropped. Shaded bands are 95% confidence intervals. Formal differential-slope tests yield $p = 0.838$ for pad use and $p = 0.459$ for the NFHS sanitary-napkin category, as reported in Section 4.2.

Data source: Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers, Government of India; National Family Health Survey Round 4; SHRUG: 2011 Population Census Abstract and Village Directory; VIIRS Night Lights.

5 Results

This section presents our findings on the impact of subsidized sanitary napkin provision through the *Jan Aushadhi Kendra* network. We begin by documenting the program’s effect on menstrual-product adoption and then examine heterogeneity by local supply-side infrastructure and owner gender. We also show that the adoption effect builds over the first six to eight months of *Suvidha* availability rather than appearing immediately.

5.1 Effect on Menstrual Health Management

Table V presents estimates of the impact of *Suvidha* availability through *Jan Aushadhi Kendras* on pad use. We implement the doubly robust difference-in-differences estimator of [Sant’Anna and Zhao \(2020\)](#), as described in Section 4. Identification draws on two sources of variation: a cross-sectional contrast between subdistricts that ever receive a *Kendra* and those that never do, and a temporal contrast based on whether a woman is interviewed before or after *Suvidha* pads become available to her. In treated subdistricts, this temporal contrast is defined by local availability; in never-treated subdistricts, which have no opening date of their own, it is anchored to the date when *Suvidha* becomes available nationally, which provides the natural counterfactual clock.

The specification includes individual-level characteristics (religion, caste, urban/rural status, and age), baseline subdistrict characteristics that may predict both *Kendra* placement and pad use, district fixed effects, and survey month-year fixed effects. Standard errors are clustered at the subdistrict level.

Table V shows that availability of *Suvidha* through *Kendras* increases the use of sanitary napkins or locally prepared napkins by 9.6 percentage points (SE = 1.1 percentage points). The estimate is statistically significant at the 1% level and is about 16% of the 60.3% comparison mean.

TABLE V: Treatment Effect of Provision of Subsidized *Suvidha* Sanitary Napkins via Jan Aushadhi *Kendras* on Pad Use

	(1)
$\hat{\tau}$	0.096 (0.011)
Mean	0.603
Obs	480,105

Note: Pad use is defined in Section 3.4. This table reports the estimate from [Sant’Anna and Zhao \(2020\)](#)’s doubly robust difference-in-differences framework. Covariates include baseline subdistrict aggregates of village amenities, female literacy, female labor force participation, and mean nightlights, as well as individual-level characteristics (religion, caste, urban/rural status, and age), district fixed effects, and survey month-year fixed effects. Standard errors are clustered at the subdistrict level and reported in parentheses.

Data sources: Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers, Government of India; National Family Health Survey Rounds 4 and 5; SHRUG; VIIRS Night Lights.

The primary outcome is the use of either a locally prepared pad or a sanitary napkin, not *Suvidha* specifically: the NFHS records product type but not brand. The estimate therefore captures the effect of subsidized availability on reported pad adoption rather than on *Suvidha* purchases alone. Appendix Table A6 distinguishes use of each product from use without the other two listed materials. Sanitary-napkin use rises by 3.9 percentage points, locally prepared pad use rises by 5.0 percentage points, and cloth use falls by 3.9 percentage points. These outcomes are nonexclusive: a woman may report using more than one material.

The exclusive-use outcomes reveal an additional change in product-use patterns. Sanitary-napkin-only use increases by 10.6 percentage points, while cloth-only use declines by 4.5 percentage points and locally prepared-pad-only use increases by 1.1 percentage points. The larger increase in sanitary-napkin-only use than in any sanitary-napkin use is consistent with a shift away from combining sanitary napkins with other materials, alongside an increase in sanitary-napkin use overall. By contrast, the smaller increase in exclusive use of locally prepared pads suggests that their increased use is predominantly alongside another listed material. Taken together, the estimates indicate changes in both adoption and product combinations, rather than one-for-one substitution of *Suvidha* pads for cloth.

Appendix Figure A2 shows that national *Suvidha* sales rose sharply after the ₹1 price took effect. Beyond direct purchases of the subsidized product, the intervention may have encouraged women to try sanitary napkins and, through that experience, develop a preference for their use. Such a preference need not be specific to *Suvidha*: women who initially try the subsidized product may subsequently purchase other brands because they prefer their characteristics or because *Suvidha* is unavailable when needed. Subsidized access could therefore encourage adoption of sanitary napkins more broadly, rather than translate one-for-one into continued use of the subsidized brand. This interpretation allows for both direct *Suvidha* adoption and increased use of non-*Suvidha* sanitary napkins, without attributing the entire estimated response to either. The NFHS records self-reported product type, but not brand, quantities purchased, or individual purchasing histories. Separating the contribution of *Suvidha* adoption from that of other brands, and establishing whether experimentation, preference formation, or supply constraints account for these choices, is therefore beyond the scope of this paper. Our estimates capture the overall response in reported product use to subsidized availability through the *Kendra* network, rather than a brand-specific demand response or a pure price elasticity.

5.2 Heterogeneous Effects

We next examine whether the adoption response differs with the local environment in which a *Kendra* operates. We focus on nearby health and retail infrastructure and on the gender of the *Kendra* owner.

To characterize the supply-side heterogeneity, we estimate the baseline specification separately across mutually exclusive subsamples defined by the characteristic of interest and compare the resulting treatment effects. For each characteristic we partition treated subdistricts into two mutually exclusive regimes—for example, subdistricts where a *Kendra* opened within five minutes’ driving distance of a hospital versus those where it opened without a nearby hospital—and estimate the baseline specification on each regime against a common comparison group of never-treated subdistricts. Treatment is whether a subdistrict ever experienced a *Kendra* opening of the relevant type, and the post indicator equals one for women interviewed after the corresponding opening; each regime excludes the complementary opening type. Appendix Figure A3 illustrates the construction for hospital proximity, and the pharmacy, supermarket, and owner-gender cuts follow the same template.

Health and Retail Infrastructure Table VI reports the estimates. Point estimates are larger where *Kendras* co-locate with existing infrastructure: 12.8 versus 7.9 percentage points near a hospital ($t = 2.10$ on the difference), 8.9 versus 6.2 near a non-generic pharmacy ($t = 1.18$), and 15.5 versus 5.0 near a supermarket ($t = 4.56$). The hospital and supermarket gaps are statistically distinguishable, whereas the pharmacy gap is not. Adoption thus responds most strongly where a *Kendra* opens amid existing health and retail infrastructure, although these comparisons may also reflect broader differences across locations.

Gender of *Kendra* Owner Columns 7–8 examine how ownership structure relates to adoption. Woman-owned *Kendras* raise pad use by 11.0 percentage points, compared with 5.0 percentage points at other outlets, a 6.0-percentage-point differential ($t = 2.41$). Several channels could contribute to this gap. Woman-owned *Kendras* may locate where gender norms differ; women entrepreneurs also receive program support, including an application-fee waiver and setup incentives; and owner gender does not identify the person serving customers at the counter. At the same time, women owners may be more likely to promote the product, provide information, or make an intimate purchase more comfortable. The owner-gender comparison is therefore suggestive of transaction and information channels but is not a clean test of stigma alone.

Effects by Time Since Availability A final source of heterogeneity is exposure duration: how long *Suvidha* pads have been available to a woman at the time she is interviewed. Table VII re-

TABLE VI: Treatment Effect of Subsidized *Suvidha* Availability on Napkin Use by Local Amenities and Owner Gender

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Hospital		Pharmacy		Supermarket		Woman-owned	
	Yes	No	Yes	No	Yes	No	Yes	No
$\hat{\tau}$	0.128 (0.012)	0.079 (0.020)	0.089 (0.019)	0.062 (0.013)	0.155 (0.019)	0.050 (0.014)	0.110 (0.021)	0.050 (0.013)
t		2.097		1.179		4.555		2.409
Mean	0.615	0.582	0.625	0.584	0.625	0.581	0.624	0.585
Obs	407,607	275,635	304,035	379,207	325,935	357,307	328,493	354,749

Note: This table reports estimates from [Sant’Anna and Zhao \(2020\)](#)’s doubly robust difference-in-differences framework for pad use. Covariates include baseline subdistrict aggregates of village amenities, female literacy, female labor force participation, and mean nightlights, as well as individual-level characteristics (religion, caste, urban/rural status, and age), district fixed effects, and survey month-year fixed effects. Standard errors are clustered at the subdistrict level and reported in parentheses. The t statistic in each pair tests equality of the two subsample treatment effects.

Columns (1) and (2) compare outcomes when a *Kendra* opened near versus not near a hospital. Columns (3) and (4) present analogous comparisons for a non-generic pharmacy; columns (5) and (6), for a supermarket; and columns (7) and (8), for whether the *Kendra* is woman-owned.

Data source: Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers, Government of India; National Family Health Survey Rounds 4 and 5; SHRUG: 2011 Population Census Abstract and Village Directory; VIIRS Night Lights. Geoapify Places API; Claude AI; dataset of Indian female names ([Bejda, 2015](#)).

estimates the baseline doubly robust specification while progressively widening the post-treatment window, restricting treated post-period observations to women interviewed within 60, 120, 240, and 360 days of *Suvidha* becoming available in their subdistrict. The estimated effect rises monotonically with exposure duration, from 3.1 percentage points within 60 days to 5.3 percentage points within 120 days, 10.3 within 240 days, and 10.4 within 360 days, each estimated precisely. The effect builds over roughly the first six to eight months of availability and then levels off; the 240- and 360-day estimates sit marginally above the full-sample estimate of 9.6 percentage points in Table V because the full sample additionally includes the longest-exposure women—interviewed latest, and thus deepest into the COVID-19 fieldwork window—among whom the estimated effect is somewhat attenuated (Section 5.4.4). This ramp-up is consistent with gradual diffusion of information about the subsidized product and with adoption occurring at successive menstrual cycles rather than immediately upon availability, a pattern in keeping with the social-learning dynamics documented for other new technologies ([BenYishay and Mobarak, 2019](#); [Conley and Udry, 2010](#)).

TABLE VII: Treatment Effect on Napkin Use by Time Since *Suvidha* Availability

	(1)	(2)	(3)	(4)
	60	120	240	360
$\hat{\tau}$	0.031*** (0.012)	0.053*** (0.012)	0.103*** (0.011)	0.104*** (0.011)
Mean	0.603	0.603	0.603	0.603
Obs	398,350	408,503	437,724	438,446

Note: This table reports estimates from Sant’Anna and Zhao (2020)’s doubly robust difference-in-differences framework for pad use. Each column re-estimates the baseline specification restricting treated post-period observations to women interviewed within 60, 120, 240, or 360 days of *Suvidha* availability; never-treated subdistricts are retained throughout. Covariates are the same as in Table V. Standard errors are clustered at the subdistrict level and reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Data source: Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers, Government of India; National Family Health Survey Rounds 4 and 5; SHRUG: 2011 Population Census Abstract and Village Directory; VIIRS Night Lights.

5.3 Mechanisms

Four empirical patterns organize the discussion. First, the main estimate indicates that subsidized *Suvidha* availability through *Jan Aushadhi Kendras* increased use of either locally prepared pads or sanitary napkins by 9.6 percentage points overall (Table V). Second, point estimates are larger near hospitals (12.8 versus 7.9 percentage points) and supermarkets (15.5 versus 5.0 percentage points). Third, the estimated response is larger at woman-owned outlets (11.0 versus 5.0 percentage points; Table VI). Fourth, estimates are larger in progressively wider post-availability windows (Table VII). These comparisons describe where estimated responses differ within our sample and motivate several possible complementary channels: improving *physical access*, relaxing an *affordability* constraint, increasing *information* about menstrual products, and reducing the transaction costs or *stigma* of obtaining them.

The infrastructure pattern is consistent with these channels reinforcing one another. Locations near hospitals and supermarkets may offer greater visibility and more opportunities for information exchange, while women already visiting these areas can combine a pad purchase with routine errands or clinical visits. Buying pads alongside other health and retail goods may also make the transaction more comfortable. The social-learning literature motivates the possibility that communication and experience influence adoption (BenYishay and Mobarak, 2019; Conley and Udry, 2010). Supply conditions may further reinforce these complementarities. Studies of *Kendra* medicine availability document stockouts and suggest that operators’ ordering decisions depend on local demand as well as distributor supply (Lavtepatil and Ghosh, 2022; Bajaj, Divya Lakshmi S., Bhat, and Kulkarni, 2024). A larger customer base could encourage operators to carry deeper inventories and order more frequently, making subsidized provision more effective where demand supports those investments. However, higher demand can also exhaust stocks more quickly. Qual-

itative reporting documents customers obtaining medicines from conventional pharmacies when *Kendras* lack stock, as well as referrals between the two channels according to availability and affordability (Kaur, 2017; Jani, 2025). Although this evidence concerns medicines rather than sanitary napkins, it motivates a potential commercial fallback mechanism: conventional retailers may complement subsidized provision by filling availability gaps, even while competing for individual purchases. Therefore, both better stocking and higher retail complementarity in higher-traffic locations could be plausible explanations for the heterogeneity.

The larger treatment effects in healthcare-dense areas, together with the wealth gradient in pad use (Table IV), motivate affordability as a potential barrier, although this mechanism cannot be isolated. Similarly, the larger point estimate at woman-owned outlets cannot be attributed to reduced stigma alone. The owner-gender comparison is consistent with both transactional and informational channels. A same-gender interaction may reduce discomfort, a possibility broadly motivated by evidence that gender concordance matters in other health-related interactions (Cabral and Dillender, 2024). Women owners may also be more active in promoting the product, providing information, or discussing menstrual health. Ownership, however, does not identify the person serving customers. Differences in local gender norms and access to program support provide further possible explanations.²

Finally, affordable access may encourage women to experiment with sanitary napkins and develop a preference for their use. That preference need not be specific to *Suvidha*: women may subsequently choose other brands because of product characteristics or because the subsidized product is unavailable. Nearby commercial retailers could facilitate such choices. Taken together, the patterns are consistent with affordability, information, transaction conditions, and supply availability operating jointly. Their relative contributions are not separately identified by this design.

5.4 Robustness Checks

We first present the dynamic event-study estimates and then conduct five checks on the main adoption result. The first exploits *Kendra* openings before subsidized pads existed. The second removes interviews from the initial ₹2.50 price regime. The third addresses the COVID-19 suspension of NFHS-5 fieldwork. The fourth applies the sensitivity analysis of Rambachan and Roth (2023). The fifth excludes states with concurrent menstrual-health programs.

²The Department of Pharmaceuticals’ *Annual Report 2020–21* describes additional setup support of ₹2 lakh for eligible women and other specified entrepreneurs: ₹1.5 lakh for furniture and fixtures and ₹0.5 lakh for computers and related equipment (Department of Pharmaceuticals, 2021, pp. 37–38). This description does not establish which outlets in our sample received the support.

5.4.1 Dynamic Treatment Effects

Panel (a) of Figure III presents dynamic treatment effects from the [Callaway and Sant’Anna \(2021\)](#) estimator, where a subdistrict’s treatment cohort is the calendar year in which *Suvidha* becomes available. The event-time-zero estimate is 0.092 (SE = 0.013), close to the main doubly robust estimate, and the available pre-treatment coefficients are non-positive.

Two features of the NFHS shape how we estimate the event study. First, fieldwork within each round was spread over multiple years, and *Kendras* were established in subdistricts on different dates, so women are observed at several event times relative to local availability rather than at two fixed points. Second, fieldwork was clustered in time within geographies: most interviews in a district occurred close together, leaving little within-district variation in interview timing. Therefore, this specification cannot support district fixed effects.

5.4.2 Falsification Test: *Kendras* Without *Suvidha* Availability

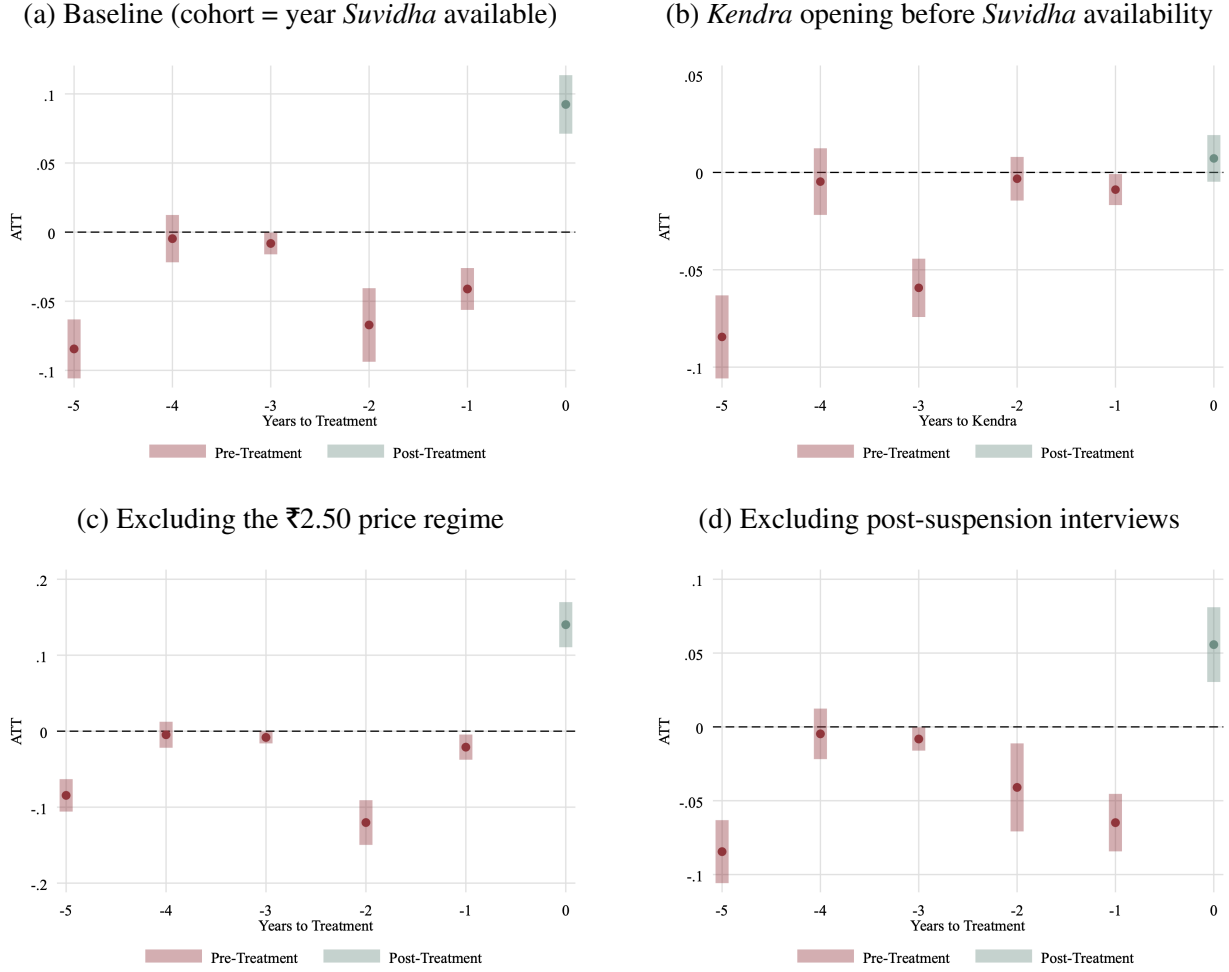
Our first check exploits the temporal gap between *Kendra* establishment and *Suvidha* pad availability. *Kendras* began operating in 2008, but subsidized *Suvidha* pads were not part of their product line until a decade later. If the adoption response in Panel (a) of Figure III reflects the subsidized product rather than some other feature of *Kendra* entry—health information, retail density, or correlated local development—then *Kendra* openings in the pre-*Suvidha* era should produce no movement in pad use.

Panel (b) of Figure III implements this test. We restrict the sample to interviews conducted entirely before *Suvidha* became available and re-estimate the [Callaway and Sant’Anna \(2021\)](#) event study, now defining a subdistrict’s treatment cohort by the calendar year its *Kendra* opens. In sharp contrast to the discrete jump at *Suvidha* availability in Panel (a) of Figure III, the estimate at *Kendra* opening is small and statistically indistinguishable from zero. *Kendra* presence alone, absent the subsidized product, does not move pad use at opening.

5.4.3 Excluding the ₹2.50 Price Regime

Suvidha pads initially retailed at ₹2.50 per pad (June 2018 onward) before the price was reduced to ₹1 (August 2019 onward). The initial ₹2.50 regime reached consumers in far smaller volume than the subsequent ₹1 rollout (as seen in Appendix Figure A2), which is why we anchor treatment to the ₹1 launch date rather than the initial launch. Women interviewed after the ₹2.50 launch but before the price reduction are coded as untreated in our baseline specification despite facing a partially subsidized product, which could attenuate the estimated effect. Panel (c) of Figure III re-estimates the event study excluding NFHS-5 interviews conducted during this transitional win-

FIGURE III: Dynamic Treatment Effects: Baseline, Falsification, and Exclusion Checks



Note: All four panels plot dynamic average treatment effects on the treated from the [Callaway and Sant'Anna \(2021\)](#) estimator with doubly robust estimation, using not-yet-treated subdistricts as controls; the outcome is the primary indicator for use of either locally prepared or sanitary napkins among women aged 15–24. Panel (a) is the baseline event study: a subdistrict's cohort is the calendar year in which *Suvidha* pads become available. In Panel (b), the sample is restricted to women interviewed before *Suvidha* pads became available, and a subdistrict's cohort is the calendar year in which its *Kendra* opens. In Panel (c), the cohort is again the year *Suvidha* becomes available, and the sample excludes NFHS-5 interviews conducted while *Suvidha* retailed at ₹2.50, before the ₹1 price took effect. In Panel (d), the sample excludes all NFHS-5 interviews conducted after March 2020, removing the fieldwork resumed after the COVID-19 suspension so that identification rests only on interviews conducted before the suspension. Covariates include individual-level characteristics (religion, caste, urban/rural status, and age) and baseline subdistrict aggregates of village amenities, female literacy, female labor force participation, and mean nightlights. Shaded bands are 90% confidence intervals.

Data source: Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers, Government of India; National Family Health Survey Rounds 4 and 5; SHRUG: 2011 Population Census Abstract and Village Directory; VIIRS Night Lights.

dow.³ The post-treatment estimate is, if anything, larger than the baseline, and the pre-period pattern is unchanged. The main effect is therefore not an artifact of the price transition; to the extent the transitional window matters, our baseline estimate is conservative.

5.4.4 COVID-19 Fieldwork Suspension and Interview Timing

NFHS-5 fieldwork was interrupted by the COVID-19 pandemic: interviewing was suspended in the spring of 2020 and resumed only several months later, leaving a multi-month gap in the field calendar. Because the ₹1 price had taken effect before the fieldwork suspension, women interviewed after fieldwork resumed had lived under subsidized availability substantially longer than those interviewed before the suspension, leaving more time for adoption to materialize. If treated (*Kendra*-served) women were disproportionately represented among these later interviews, the mechanical accumulation of exposure time—rather than the causal effect of the subsidy—could inflate the estimated effect.

Appendix Figure A4 plots the distribution of interview dates for treated and never-treated women in NFHS-5. The two distributions track each other closely throughout the pre-suspension window and both exhibit the characteristic gap during the suspension; in the resumed-fieldwork window, treated women are modestly over-represented relative to never-treated women. This imbalance motivates a direct check. Panel (d) of Figure III re-estimates the event study after dropping all interviews conducted after March 2020—the entire post-suspension window—so that identifying variation comes solely from the pre-suspension period, during which exposure time is bounded and comparable across groups. The post-treatment estimate remains positive and statistically significant, comparable to the baseline. Differential interview timing around the COVID-19 fieldwork suspension therefore does not account for the adoption effect.

5.4.5 Sensitivity to Parallel-Trends Violations

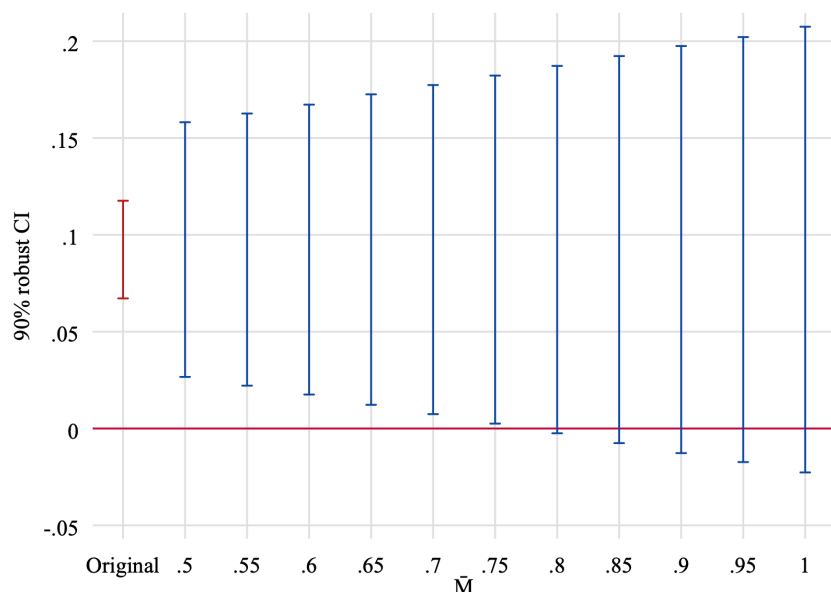
Because the pre-period of the event study in Panel (a) of Figure III exhibits some individually significant (but negative) coefficients, we assess how sensitive the adoption result is to violations of parallel trends using the sensitivity analysis of [Rambachan and Roth \(2023\)](#). We apply the relative-magnitudes restriction $\Delta^{RM}(\bar{M})$, which bounds the post-treatment violation of parallel trends by $\bar{M}$ times the largest pre-period violation, to the [Callaway and Sant’Anna \(2021\)](#) event-study estimates for pad use, and report 90% robust confidence intervals.

Figure IV presents the results. The robust confidence interval excludes zero for $\bar{M} \leq 0.75$: the positive adoption effect survives post-treatment violations of parallel trends as large as three-

³NFHS-5 interviews began in June 2019, so this restriction excludes those interviewed between June and August 2019.

quarters of the maximal pre-period deviation. The interval first includes zero at $\bar{M} = 0.80$, placing the breakdown value between $\bar{M} = 0.75$ and $\bar{M} = 0.80$ on the evaluated grid. Given that the largest pre-period deviation in Panel (a) of Figure III is itself imprecisely estimated at distant event time, we read this as meaningful robustness: overturning the result requires a post-treatment confound roughly three-quarters the size of the most extreme—and least precisely identified—pre-period fluctuation, read alongside the other supporting evidence above.

FIGURE IV: Sensitivity to Parallel-Trends Violations: Relative Magnitudes $\Delta^{RM}(\bar{M})$, Napkin Use



Note: 90% robust confidence intervals from [Rambachan and Roth \(2023\)](#) under the relative-magnitudes restriction $\Delta^{RM}(\bar{M})$, applied to the [Callaway and Sant’Anna \(2021\)](#) event-study estimates for pad use in Panel (a) of Figure III, over a grid of $\bar{M}$ from 0.5 to 1 in steps of 0.05. “Original” denotes the baseline confidence interval. The robust interval first includes zero at $\bar{M} = 0.80$.

Data source: Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers, Government of India; National Family Health Survey Rounds 4 and 5; SHRUG: 2011 Population Census Abstract and Village Directory; VIIRS Night Lights.

5.4.6 Concurrent State Menstrual Health Programs

Several states operated menstrual-hygiene programs during the study window, raising the concern that our estimate partly reflects concurrent interventions. Kerala’s *She Pad* began in November 2017, Odisha’s *Khushi* in February 2018, and Maharashtra’s *Asmita Yojana* in March 2018 ([Kerala State Women’s Development Corporation, n.d.](#); [Press Trust of India, 2018b,a](#)). Haryana introduced its scheme in August 2020 ([Haryana Directorate of Information, Public Relations and Languages, 2020](#)). Tamil Nadu’s rollout began in March 2012, while Karnataka’s *Shuchi* dates to 2013–14 ([Tamil Nadu Health and Family Welfare Department, 2016](#); [Vinobha K. T., 2023](#)). Punjab launched *Udaan* in May 2021, at the end of the stated NFHS-5 window; Andhra Pradesh’s October 2021 and

Rajasthan’s December 2021 launches occurred after that window ([Indo-Asian News Service, 2021](#); [The News Minute, 2021](#); [Rajasthan Directorate of Economics and Statistics, 2022](#)). The latter two exclusions therefore extend the geographic robustness check rather than remove exposure to these particular schemes during the stated survey period.

Appendix Table A3 summarizes the programs and their sources. They differ from *Suvidha*’s retail channel in their delivery arrangements and eligibility rules, which include school enrollment, age, residence, or household disadvantage. Kerala’s *She Pad* scheme has been shown to reduce dropout among school-going girls ([Agarwal, Chia, and Ghosh, 2024](#)). Programs overlapping fieldwork could also affect pad adoption, so correlation between their geography and *Kendra* rollout could confound the comparison.

We address this concern by re-estimating the main specification while excluding each of the nine states separately and then all nine jointly. Table VIII reports the results. Every single-state exclusion remains positive and statistically significant. Excluding all nine states simultaneously reduces the estimate to 5.3 percentage points (SE = 1.5 percentage points), with a comparison mean of 55.9%.

The smaller estimate in the joint-exclusion sample may reflect differences in the composition and timing of the excluded states as well as interactions with their own menstrual-health programs. We therefore use the joint exclusion as a robustness check rather than as evidence of complementarity. The relevant result is that the estimated effect remains positive and statistically significant in states without these concurrent programs.

TABLE VIII: Treatment Effect on Napkin Use: Excluding States with Concurrent Menstrual-Health Programs

Excluded state	$\hat{\tau}$	Mean	Obs
Punjab	0.100*** (0.012)	0.599	468,152
Haryana	0.101*** (0.012)	0.598	465,451
Rajasthan	0.103*** (0.011)	0.601	447,475
Odisha	0.095*** (0.011)	0.606	461,207
Maharashtra	0.097*** (0.011)	0.601	461,471
Andhra Pradesh	0.138*** (0.011)	0.597	465,038
Kerala	0.076*** (0.011)	0.600	474,098
Tamil Nadu	0.102*** (0.011)	0.596	465,259
Karnataka	0.075*** (0.011)	0.602	463,257
All nine excluded	0.053*** (0.015)	0.559	330,568

Note: Each row reports the estimate from Sant’Anna and Zhao (2020)’s doubly robust difference-in-differences framework, re-estimated on the sample excluding the indicated state; the final row excludes all nine states simultaneously. Covariates include baseline subdistrict aggregates of village amenities, female literacy, female labor force participation, and mean nightlights, as well as individual-level characteristics (religion, caste, urban/rural status, and age), district fixed effects, and survey month-year fixed effects. Standard errors, clustered at the subdistrict level, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Data sources: Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers, Government of India; National Family Health Survey Rounds 4 and 5; SHRUG: 2011 Population Census Abstract and Village Directory; VIIRS Night Lights.

Where to Expand: A Planner’s Siting Problem

The same heterogeneity that complicates the mechanism story has a clear implication for where to grow the network, exposing a tension between efficiency and equity. Because adoption responds most strongly where *Kendras* co-locate with existing infrastructure, a planner maximizing aggregate adoption per rupee would concentrate openings in already well-served regions—yet these are not the areas of greatest deprivation. We formalize the tradeoff as a constrained location-choice problem. Let $j = 1, \dots, N$ index subdistricts without a *Kendra*, each with n_j women of menstruating age, a predicted adoption gain $\hat{\tau}_j \in [0, 1]$ —obtained by mapping the heterogeneous effects of

Section 5.2 to subdistrict characteristics—and an indicator $d_j \in \{0, 1\}$ equal to one if the subdistrict is a healthcare desert. The planner chooses a set S of K subdistricts:

$$\max_{S \subseteq \{1, \dots, N\}, |S|=K} \sum_{j \in S} n_j [(1 - \lambda) \hat{\tau}_j + \lambda d_j], \quad (4)$$

where $\lambda \in [0, 1]$ is the equity weight: at $\lambda = 0$, the planner maximizes aggregate predicted adoption; at $\lambda = 1$, the planner maximizes the underserved population reached. Because the objective is additive, the solution ranks subdistricts by $s_j(\lambda) = n_j[(1 - \lambda)\hat{\tau}_j + \lambda d_j]$ and takes the top K . The tension enters through the negative association between $\hat{\tau}_j$ and d_j : the underserved subdistricts where unmet need is greatest have the smaller predicted gains, since adoption responds most strongly where *Kendras* co-locate with existing infrastructure. Varying λ therefore traces an efficiency–equity frontier rather than reordering a stable ranking. An interactive tool⁴ allows users to explore the allocation problem in (4) for user-chosen values of K and λ : openings concentrate in well-connected urban subdistricts when λ is low and disperse into underserved regions as λ rises. The choice of expansion rule thus embeds an implicit distributional judgment rather than a purely technical optimization.

5.5 Health Outcomes

The adoption results establish that subsidized availability changes menstrual-product use. We next ask whether the symptom measures available in the NFHS move in a direction consistent with improved menstrual health. A health effect is plausible if disposable napkins replace poorly washed, incompletely dried, or repeatedly reused cloth, or make it easier to change absorbent material more frequently. Earlier work associates unhygienic cloth use with reproductive-tract-infection symptoms, and experimental evidence finds that provision of menstrual products can affect some infection outcomes, although the evidence for sanitary pads specifically is mixed (Phillips-Howard, Nyothach, ter Kuile, Omoto, Wang, Zeh, Onyango, Mason, Alexander, Odhiambo, Eleveld, Mohammed, van Eijk, Edwards, Vulule, Faragher, and Laserson, 2016). The outcomes available here are much less specific: the NFHS asks a subset of respondents whether they experienced genital discharge or a genital sore or ulcer during the preceding 12 months. They are self-reported symptoms, not clinical diagnoses.

Table IX reports estimates for these outcomes. The health questions are available for only about 79,000 of the 489,013 women in the descriptive sample, so these estimates are based on a much smaller survey-eligible subsample than the pad-use results. In the full survey-eligible sample, the estimates are mixed: any genital symptom rises by 1.4 percentage points (SE = 0.4 percentage

⁴Available at <https://mishradibya.com/pmbjp/>.

points), discharge rises by 2.9 percentage points (SE = 0.4 percentage points), and sore or ulcer falls by 0.8 percentage points (SE = 0.3 percentage points).

Panel B excludes women who are pregnant at interview, since pregnancy can itself affect genital discharge and related symptoms. All three estimates are negative in this sample: -2.7 percentage points for any genital symptom (SE = 0.4 percentage points), -1.6 percentage points for discharge (SE = 0.4 percentage points), and -0.9 percentage points for sore or ulcer (SE = 0.2 percentage points). The pattern is consistent with a possible health benefit of shifting menstrual management toward napkins, but the difference between the full and non-pregnant samples makes the result sensitive to sample definition. We therefore treat these estimates as suggestive rather than as a second main causal result.

TABLE IX: Treatment Effect of Subsidized *Suvidha* Availability on Health Outcomes

	(1)	(2)	(3)
	Any genital symptom	Genital discharge	Genital sore/ulcer
<i>Panel A: Full survey-eligible sample</i>			
$\hat{\tau}$	0.014*** (0.004)	0.029*** (0.004)	-0.008 *** (0.003)
Mean	0.044	0.039	0.014
Obs	77,330	77,352	77,362
<i>Panel B: Non-pregnant women</i>			
$\hat{\tau}$	-0.027 *** (0.004)	-0.016 *** (0.004)	-0.009 *** (0.002)
Mean	0.039	0.035	0.013
Obs	72,455	72,475	72,484

Note: Each column re-estimates the doubly robust difference-in-differences specification with the indicated binary genital-symptom outcome. “Any genital symptom” equals one if the respondent reports either genital discharge or a genital sore/ulcer. Panel A uses respondents with valid yes/no responses to the relevant symptom questions; Panel B additionally excludes women who are pregnant at interview. The row labeled “Mean” reports the comparison-group mean in the corresponding estimation sample. The symptom questions refer to the preceding 12 months and are self-reported rather than clinical diagnoses. Covariates are the same as in Table V; standard errors are clustered at the subdistrict level. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Furthermore, the symptom questions refer to the preceding 12 months. For a woman interviewed only a few months after *Suvidha* becomes available, part of that recall period therefore predates treatment. If access reduces symptoms only after it begins and does not change reporting, this timing mismatch will attenuate measured improvements for recently treated women. The estimates are consequently effects on a rolling 12-month symptom measure, not effects of a full year of exposure.

The dynamic evidence is also less clean than for pad use. Appendix Figure A5 plots the three health event studies separately for non-pregnant women. Event-time-zero estimates are negative for all three outcomes, but the pre-treatment pattern is not uniformly flat: the average pre-treatment

effect is statistically insignificant for the composite and sore/ulcer outcomes and marginal at the 10% level for discharge, with some individual pre-treatment coefficients also statistically significant. Additionally, Appendix Table A7 reports the pre-*Suvidha* falsification test. Re-anchoring treatment to pre-*Suvidha Kendra* openings yields insignificant placebo effects for the composite and sore/ulcer outcomes among non-pregnant women, but a positive and significant placebo effect for discharge, which is reassuring evidence that the negative effect reported in Table IX is specific to the availability of *Suvidha* and not merely the presence of *Kendras*.

6 Conclusion

This paper asks whether subsidizing sanitary napkins and delivering them through a network of generic pharmacies raises their adoption, exploiting India’s *Jan Aushadhi–Suvidha* program. Linking administrative records on *Kendras* to two rounds of the National Family Health Survey, we find that deeply subsidized availability raises pad use by 9.6 percentage points, a 16% increase over a 60.3% comparison mean. The adoption estimate remains positive in the parallel-trends sensitivity analysis, timing checks, and exclusions of states with concurrent menstrual-health programs. The pre-*Suvidha Kendra*-opening falsification yields no comparable adoption response.

These findings speak to several literatures. Most of what is known about menstrual-product interventions comes from experiments that provide products directly and trace downstream effects on schooling, health, and labor; we estimate instead the first-order behavioral response—whether deeply subsidized provision draws prior non-users into pad use—at national scale and through a retail channel rather than schools or frontline health workers. In doing so we extend the literature on inexpensive welfare-enhancing goods beyond one-off and preventive products to a recurring necessity, and we add evidence on menstrual-product adoption in a setting where baseline use is far from universal. Because *Suvidha* changes both the price of pads and their availability through the *Kendra* network, the estimate should not be interpreted as a pure price elasticity.

On the supply side, adoption rises most where a *Kendra* opens amid existing health and retail infrastructure and at woman-owned outlets. These patterns show where the response is largest but cannot cleanly separate the roles of information, stigma, affordability, inventory, and physical access. They also point to an efficiency–equity tension for further expansion: the locations with the largest estimated adoption response are not necessarily those with the greatest unmet need.

The health outcomes provide more tentative evidence. Results are mixed in the full survey-eligible sample, while all three symptom estimates are negative among women who are not pregnant at the time of the interview. These outcomes are available for a much smaller NFHS subsample, refer to the preceding 12 months, and do not exhibit uniformly clean pre-treatment patterns; the pre-*Suvidha Kendra*-opening falsification is also not null for every outcome. We therefore view the

symptom estimates as suggestive evidence on downstream health rather than as a definitive effect of sustained exposure.

This analysis motivates several directions for future research. First, an important next step is to examine whether changes in menstrual-product use translate into changes in school attendance, learning, educational attainment, labor-force participation, earnings, and other longer-term human-capital outcomes. Second, studying the equilibrium interaction between *Jan Aushadhi Kendras* and conventional pharmacies or other sanitary-napkin retailers would clarify the broader market effects of subsidized provision. A central question is whether *Kendras* primarily divert purchases from commercial suppliers or expand demand for sanitary napkins by encouraging experimentation and the development of preferences that extend beyond the subsidized brand. Such a framework should also account for the joint determination of supply and demand. Operators' decisions about where to open, which products to carry, and how much inventory to maintain may depend on expected local demand, while consumers' willingness to try and continue using sanitary napkins may depend on outlet proximity, prices, and reliable availability.

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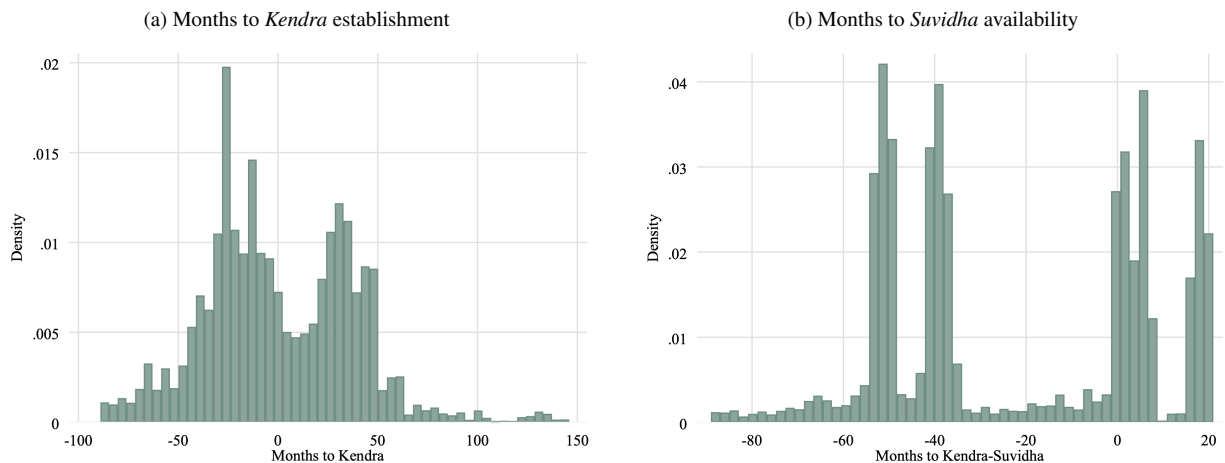
Appendix

Pennies for Pads: Subsidized Menstrual Health Products and Adoption in India

Additional Figures and Tables

Appendix Figures

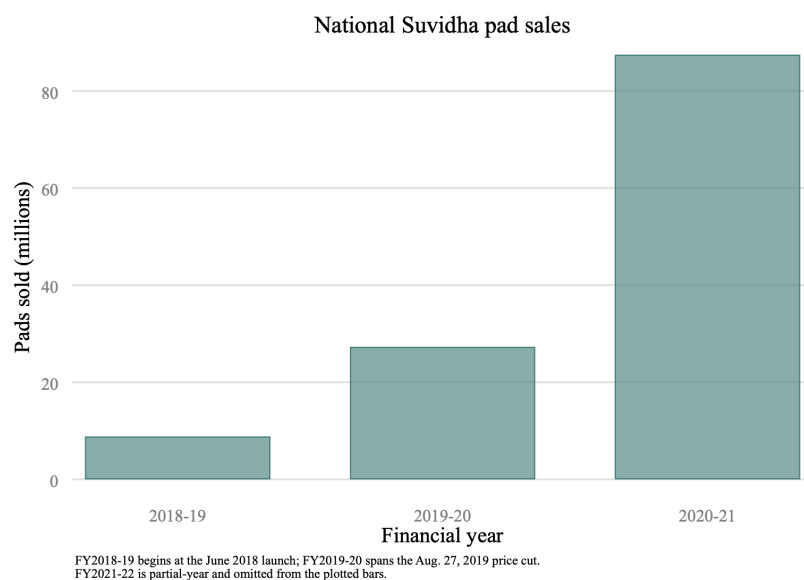
FIGURE A1: Distribution of Interview Timing Relative to *Kendra* Establishment and *Suvidha* Availability



Note: Histograms of the gap, in months, between a woman's NFHS interview month ($y_m(v007, v006)$) and (a) the establishment date of the *Jan Aushadhi Kendra* serving her subdistrict and (b) the date *Suvidha* pads became available there. Negative values denote women interviewed before the reference date; positive values, after. In panel (a) the distribution spans roughly -90 to $+150$ months, reflecting *Kendras* established years before the survey windows; in panel (b) the right tail is bounded near $+20$ months because *Suvidha* availability falls late in the panel. The multimodal shape is consistent with pooling the NFHS-4 (2015–16) and NFHS-5 (2019–21) fielding windows.

Data source: National Family Health Survey Rounds 4 and 5; *Jan Aushadhi Kendra* administrative records.

FIGURE A2: National *Suvidha* Pad Sales



Note: National units sold, aggregating the state-by-period administrative sales series. FY2018–19 begins with the June 2018 launch at ₹2.50 per pad; FY2019–20 spans the ₹1 price reduction.

Data source: Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers, Government of India.

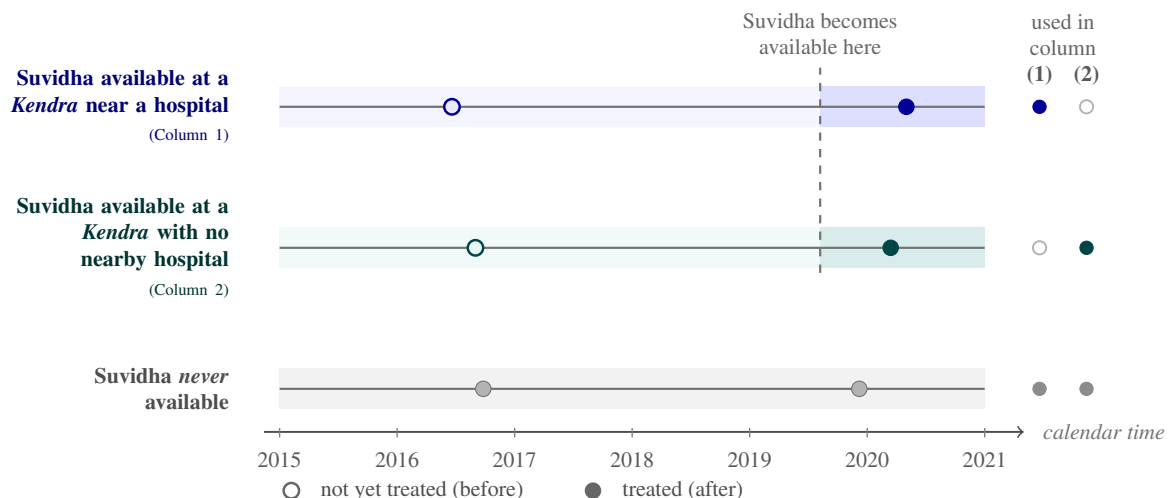
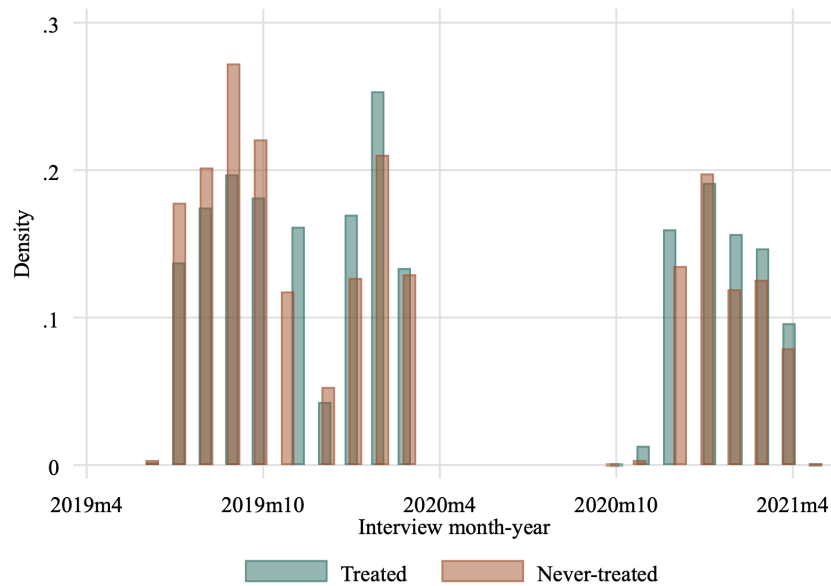


FIGURE A3: **Heterogeneity subsamples (hospital proximity).** Each lane is a representative sub-district and each marker a single interview. Among subdistricts where Suvidha becomes available at a *Kendra*, the top lane holds openings near a hospital and the middle lane openings with no hospital nearby; within each, women interviewed before the local availability date are not yet treated and those interviewed after it are treated. The bottom lane—subdistricts where Suvidha never becomes available—is the common comparison group. As the right-hand indicator shows, column (1) of Table VI is estimated on the near-hospital lane against the never-available lane, and column (2) on the no-hospital lane against the same never-available lane; each column excludes the other opening type. The pharmacy, supermarket, and owner-gender cuts follow the same template.

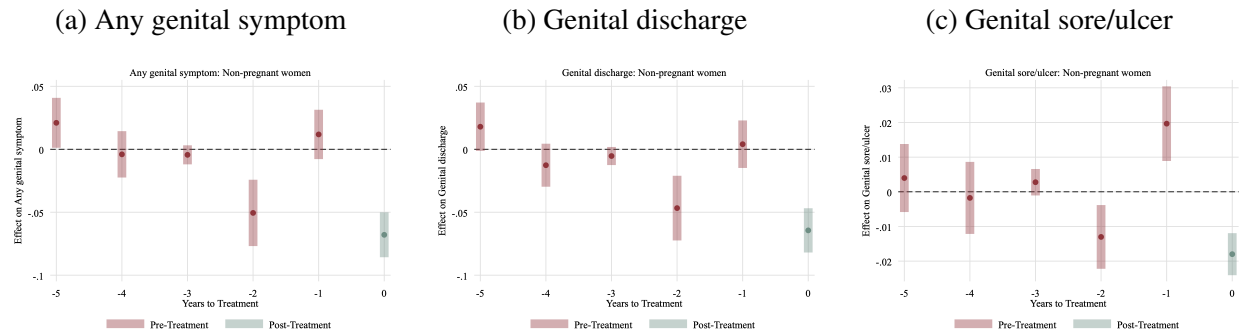
FIGURE A4: Distribution of Interview Dates by Treatment Status, NFHS-5



Note: Histograms of interview month-year for women aged 15–24 in NFHS-5, plotted separately for treated (*Kendra*-served) and never-treated subdistricts. The gap in both distributions corresponds to the suspension of NFHS-5 fieldwork during the COVID-19 pandemic. Treatment status is defined by *Kendra* service (TREAT_suvidha_2019), independent of interview date. Density is normalized within each group.

Data source: National Family Health Survey Round 5.

FIGURE A5: Dynamic Effects on Health Outcomes Among Non-Pregnant Women



Note: Each panel plots dynamic average treatment effects from the [Callaway and Sant'Anna \(2021\)](#) estimator with doubly robust estimation and not-yet-treated subdistricts as controls, using the same event-study control set and treatment-cohort definition as the dynamic specification in Section 5.4.1. The sample is restricted to non-pregnant women with valid responses to the relevant symptom question. Outcomes are self-reported symptoms during the preceding 12 months. Confidence intervals are 90%. Because the outcome recall window can partly predate treatment for recently treated women, the event-time-zero estimates should not be interpreted as the effect of a full year of exposure.

Appendix Tables

TABLE A1: Program Reach, Kendra Intensity, and Treatment-Timing Concentration

<i>Panel A: Geographic reach in linked NFHS data</i>	
Statistic	Value
Linked subdistricts	5,992
Ever-treated subdistricts	2,206
Share ever treated	0.368
<i>Panel B: Kendras per treated subdistrict</i>	
Treated subdistricts matched to Kendra registry	2,206
Mean Kendras	3.73
Standard deviation	7.65
25th percentile	1
Median	2
75th percentile	3
90th percentile	7
95th percentile	13
Maximum	163
<i>Panel C: Treatment-date concentration</i>	
Treated subdistricts	2,206
Treatment date = Aug. 27, 2019	1,705
Later local availability date	501
Share at common national date	0.773

Note: Panels A and C use unique subdistricts in the linked NFHS analysis file. Panel B merges treated subdistricts to the administrative *Kendra* registry; all 2,206 treated subdistricts match. Treatment begins on the national *Suvidha* availability date when a *Kendra* was already open and on the later local opening date otherwise.

TABLE A2: Menstrual Products Available at *Jan Aushadhi Kendras*

Drug Code	Generic Name	Unit Size	Retail Price (₹)
8140	Regular Size with Wings	Pack of 4	4.00
8141	Large Size with Wings	Pack of 4	12.00
8142	Extra Large Size with Wings	Pack of 4	15.00
8145	Regular Size with Wings	Pack of 10	10.00

Note: Prices are current retail prices in Indian rupees as set by the Department of Pharmaceuticals.

Data source: Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers, Government of India.

TABLE A3: State Sanitary Napkin Distribution Programs, 2011–2021

State	Scheme	Launch	Target population	Delivery channel and price
Tamil Nadu ^a	Menstrual Hygiene Programme	Mar. 2012	Rural girls aged 10–19, in or out of school; postnatal mothers	Free; rural adolescent outreach and government health facilities
Karnataka ^b	Shuchi	2013–14	Schoolgirls aged 10–18	Free; school distribution; interrupted c. 2020–23
Kerala ^c	She Pad	Nov. 2017	Girls in government and aided schools	Free napkins, storage, and incinerators in schools
Odisha ^d	Khushi	Feb. 2018	Girls in classes 6–12 in government and aided schools (about 1.7 million)	Free school distribution; subsidized provision to rural women also announced
Maharashtra ^e	Asmita Yojana	Mar. 2018	Rural schoolgirls and women	Subsidized; ₹5 per pack for schoolgirls; distribution through self-help groups
Haryana ^f	Mahila Avam Kishori Samman Yojana	Aug. 2020	Below-poverty-line women and girls	Free; monthly sanitary-napkin provision
Punjab ^g	Udaan	May 2021	Out-of-school girls and disadvantaged women, including BPL, slum-resident, and homeless women	Free; through <i>anganwadi</i> centers
Andhra Pradesh ^h	Swechha	Oct. 2021	Girls in classes 7–12 in government institutions (about 1 million)	Free; 10 napkins per month through schools
Rajasthan ⁱ	Udaan	Dec. 2021	Women and girls of reproductive age	Free; phased statewide distribution with self-help-group participation

Note: Dates refer to the launch or rollout identified in the cited source, not necessarily the first administrative approval. Target populations and delivery arrangements summarize the cited period; they are not measures of realized coverage. Andhra Pradesh and Rajasthan launched the listed schemes after the stated June 2019–May 2021 NFHS-5 window. BPL denotes below the poverty line.

Sources by row: (a) [Tamil Nadu Health and Family Welfare Department \(2016, Section 4.12\)](#); (b) [Vinobha K. T. \(2023\)](#); (c) [Kerala State Women's Development Corporation \(n.d.\)](#); (d) [Press Trust of India \(2018b\)](#); (e) [Press Trust of India \(2018a\)](#); (f) [Haryana Directorate of Information, Public Relations and Languages \(2020\)](#); (g) [Indo-Asian News Service \(2021\)](#); (h) [The News Minute \(2021\)](#); (i) [Rajasthan Directorate of Economics and Statistics \(2022, pp. 192–193\)](#).

TABLE A4: Descriptive Statistics: Menstrual Hygiene Management Methods Among Women Aged 15–24

	(1)	(2)
	NFHS-4	NFHS-5
Woman uses locally prepared or sanitary napkins	0.548 (0.498)	0.746 (0.435)
Woman uses cloth	0.657 (0.475)	0.519 (0.500)
Woman uses locally prepared napkins	0.150 (0.357)	0.147 (0.354)
Woman uses sanitary napkins	0.418 (0.493)	0.631 (0.482)
Observations	247,833	241,180

Note: Means and standard deviations of menstrual hygiene practices among valid NFHS respondents by survey wave. Pad use equals one for use of either locally prepared or sanitary napkins.

Data source: National Family Health Survey Rounds 4 and 5.

TABLE A5: Pre-Policy Characteristics of Never-Treated and Ever-Treated Subdistricts

	(1)	(2)	(3)
	Never treated	Ever treated	Difference
Urban share	0.113	0.300	0.187***
Mean age	19.475	19.502	0.027
Mean years of education	8.248	9.307	1.060***
Mean wealth index	-2.8e+04	1.4e+04	4.2e+04***
Community waste collection	0.020	0.069	0.049***
No waste-disposal system	0.652	0.655	0.002
Total Sanitation Campaign	0.286	0.322	0.035***
Anganwadi available	0.849	0.883	0.033***
Government school available	0.469	0.526	0.057***
Female literacy share	0.492	0.571	0.079***
Female work share	0.347	0.268	-0.079***
Nighttime lights	0.588	2.036	1.449***
Subdistricts	2,866	2,085	

Note: NFHS-4 pre-policy subdistrict means. Difference is ever treated minus never treated. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

TABLE A6: Additional Menstrual-Product Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)
	Cloth	Pad	Napkin	Only cloth	Only pad	Only napkin
$\hat{\tau}$	-0.039*** (0.007)	0.050*** (0.006)	0.039*** (0.011)	-0.045*** (0.008)	0.011*** (0.004)	0.106*** (0.009)
Mean	0.621	0.146	0.481	0.376	0.066	0.276
Obs	480,105	480,105	480,105	480,105	480,105	480,105

Note: Each column re-estimates the doubly robust specification with the indicated NFHS menstrual-product outcome. Columns 1–3 report use of cloth, locally prepared pads, and sanitary napkins, respectively. Columns 4–6 report use of each respective material without either of the other two listed materials. These exclusive-use indicators are defined only relative to these three categories. Means are comparison-sample means. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

TABLE A7: Pre-*Suvidha Kendra*-Opening Falsification for Health Outcomes

	Any genital symptom	Genital discharge	Genital sore/ulcer
<i>Panel A: Full survey-eligible sample</i>			
ATT	0.009 (0.007)	0.014** (0.007)	−0.009** (0.004)
Obs.	40,187	40,197	40,206
<i>Panel B: Non-pregnant women</i>			
ATT	0.010 (0.007)	0.014** (0.006)	−0.003 (0.004)
Obs.	37,369	37,377	37,386

Note: The sample is restricted to interviews before June 2018, before subsidized *Suvidha* pads were available. Treatment cohorts are defined by the calendar year in which a *Kendra* opened, and estimates use the [Callaway and Sant'Anna \(2021\)](#) doubly robust estimator with not-yet-treated units as controls. The table reports the simple average treatment effect on the treated. Panel B additionally excludes women who are pregnant at interview. *, **, and *** denote significance at the 10%, 5%, and 1% levels.